Elliot D'Alessandro

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EDUCATION

University of California, Berkeley **Berkeley, CA**
B.A. in Data Science and B.A. in Physics *2026–2030 (Expected)*
Coursework: Data Structures*, Discrete Math & Probability*, Structure & Interpretation of Computer Programs, Foundations of Data Science, Multivariable Calculus*, Electromagnetism & Optics*, Mechanics & Relativity
(*Spring 2027)

RESEARCH

Theoretical Computer Science Research Apprentice **Portland, OR**
Portland State University — Dr. Shravas Rao *2024*

- Worked through proofs and dual polynomial constructions in weekly faculty sessions on the open question of whether approximate degree is multiplicative under composition.
- Studied query complexity of Boolean functions across deterministic, randomized, and quantum models, including polynomial and approximate degree.
- Synthesized the current state of the problem into a formal end-of-program talk delivered to a cohort of 40 interns.

PROJECTS

Poker Bot Tournament Platform
Personal Project *2025–2026*

- Built and deployed a full-stack web platform in Flask where users submit Python poker bots that compete in automated multi-table Texas Hold'em tournaments, with hand-by-hand play streamed live to a JavaScript front end over server-sent events.
- Implemented the poker engine from scratch in Python. Handled hand evaluation and side-pot resolution, then extended it to freeze-out tournaments with escalating blinds and automatic table consolidation.
- Designed a submission pipeline for untrusted Python bots with a moderated approval queue, per-action timeouts with automatic disqualification, and encrypted storage of submitted code.
- Deployed on Linux using Nginx, Gunicorn, and systemd. Included session-based admin authentication with IP rate limiting and automated daily backups.
- Ran live tournaments with real-time statistics monitoring, Elo-based bot rankings, and leaderboard prizes, simulating real-world mechanics.

EXPERIENCE

Game Developer and Gameplay Systems Programmer **Portland, OR**
Fractal Development (Student Game Studio) *2022–2026*

- Co-founded and grew a student development team from three to seven members, directing all Unity and C# gameplay systems programming and audio composition.
- Designed and developed three games over three years for the Oregon Game Project Challenge (50+ high school teams annually), winning Best in Show, Player's Choice, Game Design, and Programming, becoming the first team to win the top two awards in one year in competition history.
- Built three games for the world's largest game jam, the GMTK Game Jam, placing in the top 2% out of 10,500+ entries (2026).
- Prototyped a Flutter mobile app consolidating schedule, room location, and grade data into one view, with the interface designed in Figma.
- Coordinated development across a seven-person team using Jira and Confluence, tracking deliverables and deadlines across annual development cycles.

Technical Camp Counselor **Hillsboro, OR**
Code Ninjas *Summer 2026*

- Taught coding and STEM curriculum to campers ages 7–14 across Scratch, JavaScript, Roblox/Lua, and Minecraft modding.
- Individually guided students through hands-on projects, managing classroom flow and engagement across multiple skill levels simultaneously.

Course Development and Teaching Assistant **Portland, OR**
Ida B. Wells High School *2025–2026*

- Supported instruction for a class of 30+ students by building curriculum, assignments, and lesson plans. Presented lectures and built demonstrations for class showcases.
- Worked one-on-one with students to explain concepts and debug code, adapting explanations for each student.

SKILLS

Languages: Python, Java, JavaScript, C#, C++, Bash, HTML/CSS
Libraries & Frameworks: pandas, NumPy, Matplotlib, Flask, Unity, Flutter
Tools: Git, Linux, Nginx, LaTeX, Figma, Blender, FL Studio, Confluence, Jira